Rating Plausibility of Word Senses in Ambiguous Sentences through Narrative Understanding

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***Abstract*—This work describes our participation in SemEval-2026 Task 5: Rating Plausibility of Word Senses in Ambiguous Sentences through Narrative Understanding. The task requires systems to predict human-perceived plausibility scores (1–5) for specific word senses based on their fit within a five-sentence narrative. We investigated the impact of narrative framing using the microsoft/deberta-v3-base model across three distinct strategies: a Simple Prompt utilizing structural concatenation, an Instructional Prompt posing a natural language question, and a Masked Prompt which replaces the target homonym with a [MASK] token. Our research demonstrates that the Simple Prompt and Masked Prompt strategies were the most effective; the Simple Prompt achieved the highest Spearman correlation ($\rho = 0.5372$), while the Masked Prompt yielded the lowest Mean Absolute Error (MAE) of 0.8332. In contrast, an Instructional Prompt and a hybrid ensemble approach—combining DeBERTa with a 4-bit quantized Mistral-7B—underperformed, with the ensemble yielding a correlation of only 0.0332. Upon evaluation on the official test dataset, our best-performing configuration achieved a Spearman correlation of 0.65, achieved by the Simple Prompt model. Our findings suggest that while simple structural prompting and masking strategies are highly effective for encoder-only architectures, the gap between validation and test performance indicates that these models generalize well to narrative nuances when prompted to prioritize context over surface-level cues.**

***Index Terms*—component, formatting, style, styling, insert**

I. Introduction

Traditionally, computers treat Word Sense Disambiguation (WSD) like a multiple-choice test where only one answer is right. However, real language is often messy and subjective. A word can have several meanings that all make sense depending on the story around it and how a reader interprets it. Because of this, human judgment is usually more of a scale than a simple "yes" or "no."

This paper describes our work for SemEval 2026 Task 5, which uses a dataset called AmbiStory to study these human intuitions. The dataset is made of five-sentence stories. The fourth sentence contains a "homonym"—a word with two different possible meanings. To understand which meaning fits, you have to look at the three sentences that come before it (the precontext) and an optional ending sentence that usually hints at which direction the story is going.

The goal of this task is to predict how plausible a specific meaning is on a scale of 1 to 5, based on ratings from real human annotators. Instead of just picking a definition, our system has to "score" how well a meaning fits the narrative logic of the story.

In our approach, we believe that an encoder-only model is the best tool for this job. While many people use generative models (like GPT or Mistral) for everything today, those models are designed to predict the next word, which can sometimes lead to "hallucinations" or irrelevant answers. In contrast, encoder-only models like DeBERTa are built to look at the whole text at once and understand the relationships between words. We argue that because this task is about judging a fit rather than writing a story, an encoder-only architecture is more reliable for catching the small narrative details that determine plausibility.

II. Evaluation Metrics

To understand how well our model performs, we need to measure the distance between the scores it predicts and the actual ratings given by humans. In this task, we primarily rely on two types of measurement: Spearman Correlation and error-based metrics (MAE and RMSE).

A. Spearman Rank Correlation (ρ)

The most important metric for this task is the Spearman correlation. Unlike standard accuracy, which only cares if the model gets the number exactly right, Spearman looks at the ranking.

In the AmbiStory dataset, some meanings are clearly better than others. We want our model to "rank" the plausible meanings higher than the implausible ones. Spearman's correlation measures the strength and direction of the relationship between these two ranked lists. It is calculated using the following formula:

$$\rho = 1 - \frac{6 \sum d_i^2}{n(n^2 - 1)}$$

Where:

- d_i is the difference between the rank of the human score and the rank of the model's predicted score for a specific item.
- n is the total number of samples.

A score of 1.0 means the model ranked everything perfectly in the same order as humans, while a score near 0 means the model is essentially guessing randomly.

B. Error Metrics: MAE and RMSE

While Spearman tells us about the "order" of the scores, we also need to know how far off our specific predictions are. For this, we use two types of error measurement:

1) **Mean Absolute Error (MAE):** This is the simplest way to measure error. We take the difference between the predicted score and the true score, and average those differences.

$$\mathrm{MAE} = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i|$$

2) **Root Mean Square Error (RMSE):** This is similar to MAE, but it squares the differences before averaging them. This makes it much more sensitive to "big misses." If the model is off by a lot on a few examples, the RMSE will increase significantly.

$$\mathrm{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}$$

C. Why Not Standard Accuracy?

In many student projects, we use "Accuracy" (the percentage of correct guesses). However, in a task like plausibility rating, accuracy is a poor fit. If the human score is 4.2 and the model predicts 4.1, a standard accuracy metric would call that a "fail" because the numbers don't match exactly. By using Spearman and MAE, we give the model credit for being "close enough" and for understanding the general narrative trend.

$$\mathrm{Correct}_i = \begin{cases} 1 & \text{if } |y_i^{\text{pred}} - \bar{y}_i^{\text{human}}| \leq \max(1, \sigma_i), \\ 0 & \text{otherwise.} \end{cases} \quad (1)$$

The overall Accuracy Within Standard Deviation is then calculated as the proportion of correct predictions over all n samples:

$$\text{Accuracy Within SD} = \frac{1}{n}\sum_{i=1}^{n}\mathrm{Correct}_i. \quad (2)$$

III. Related Work

For a long time, the NLP community treated Word Sense Disambiguation (WSD) as a simple classification task: given a word like "bank," the model just had to pick between "river bank" or "financial bank" [2]. However, as research has evolved, we have realized that meaning isn't always a "one or the other" choice.

The AmbiStory dataset, which this task is based on, builds on recent trends that look at "graded" word sense—the idea that multiple meanings can be plausible at the same time depending on how much context you have [1]. This is similar to the Word in Context (WiC) benchmark, which moved away from giant dictionaries and started looking at how words behave in specific sentence pairs [3].

Our work specifically focuses on Narrative Understanding. In the past, models mostly looked at the words immediately next to the target to find its meaning. But as studies in story logic show, sometimes the clue for a word in sentence 4 actually appears back in sentence 1 [4].

Finally, there is a big debate right now between using Generative Models (like Mistral or GPT) versus Encoder models (like DeBERTa). While generative models are great at writing, recent probes have shown that encoder-only models are often much better at "separating" different word senses in their internal layers [5]. Our decision to use a masked-input strategy with DeBERTa follows this logic: by hiding the word itself, we force the encoder to use its "narrative muscles" to reconstruct the meaning from the surrounding story [6].

IV. Method Description

In this section, we describe the architecture of our system and the different ways we formatted the input data to help the model understand narrative plausibility.

A. Model Selection

We chose `microsoft/deberta-v3-large` as our primary backbone. DeBERTa (Decoding-enhanced BERT with disentangled attention) improves upon the original BERT architecture by using a disentangled attention mechanism, which makes it particularly good at understanding the relationship between positions of words in a story. We used the `AutoTokenizer` with the `use_fast=False` setting to ensure maximum compatibility with the DeBERTa-v3 subword vocabulary.

B. Input Representations

The way a story is presented to the model (the "prompt") changes how the model "thinks." we tested three distinct formats:

1) **Rich Input:** This was a structural approach. We concatenated the context, the target sentence, and the story ending, separated by the `[SEP]` token, followed by the candidate meaning.
2) **Instructional Prompt:** We framed the task as a question-answering problem. We asked: "On a scale of 1 to 5, how plausible is the following meaning for the word '[target]'?"
3) **Masked Prompt (Our Best Performing Method):** We believed that the model might be biased by the target word itself. To fix this, we replaced the homonym with a `[MASK]` token. If the word started with a capital letter, our script automatically handled the capitalization to ensure the mask was placed correctly. We then asked the model to rate the plausibility of a meaning for that "missing" word.

C. The Ensemble Approach

We also experimented with a hybrid model. This setup combined the DeBERTa encoder with a **Mistral-7B** generative model. To keep the hardware requirements low, we loaded Mistral in 4-bit quantization.

The final prediction was a weighted average of the two models:

$$\text{Final Score} = 0.7 \times \text{DeBERTa} + 0.3 \times \text{Mistral}$$

The idea was to let DeBERTa handle the structural logic while Mistral provided general "common sense" knowledge. However, as we will discuss in the results, this combination did not perform as well as the standalone masked DeBERTa model.

D. Training Details

Training large-scale models like DeBERTa-v3-large and Mistral-7B requires significant computational power. We primarily used Google Colab's cloud GPUs for our initial experiments. However, we encountered a common hurdle: GPU resource limits. When our Colab allocation ran out, we transitioned our workflow to Kaggle, utilizing their dual T4 GPU (x2) setup.

Moving between platforms presented a few challenges. The training speed varied depending on the available hardware, and we had to be very careful with our "compute budget." This is a major reason why we limited our fine-tuning to 10 epochs. While we wanted to train for longer to see if the model would improve, the time limits on free cloud platforms meant we had to find a balance between model depth and available clock time. This resource-conscious approach is a reality for many student projects, and it forced us to focus more on smart prompting strategies (like masking) rather than just brute-force training.

V. Ablation Study

Our ablation study demonstrates the critical impact of prompt structure and target word masking on the model's ability to evaluate semantic plausibility. By incrementally adding task instructions and modifying how the target homonym is presented, we observed a steady improvement in the primary metric, Spearman's Rank Correlation (ρ), moving from a baseline of 0.0332 to a peak of 0.5372.

TABLE I
Performance comparison across different prompting strategies.

Configuration	Spearman (ρ)	MAE	RMSE
Simple Prompt	**0.5372**	**0.8930**	**1.1338**
Instructional Prompt	0.3977	0.9196	1.0937
Masked Prompt	0.5163	0.8332	1.0446
Ensemble	0.0332	1.0873	1.2867

a) Simple vs. Masked Strategy: The Simple Prompt proved most effective for capturing the relative ranking of senses. However, the Masked Prompt's success in minimizing absolute error suggests that hiding the surface form of the word forces the model to synthesize narrative clues more effectively, leading to more "cautious" and accurate score assignments.

b) The Role of Token Masking (Instructional vs. Masked): One of the most significant performance gain was achieved through the *Masked Prompt* strategy. By replacing the target homonym within the sentence with a `[MASK]` token, the Spearman correlation increased from 0.3977 to 0.5163, while the Mean Absolute Error (MAE) dropped to its lowest point (0.8332). This suggests that when the target word is visible, the model may suffer from "identity bias," where it relies too heavily on the surface form of the word. Masking the word forces the model to rely entirely on the surrounding narrative context to determine if the provided meaning is a viable semantic fit. This approach better mimics the cognitive process of disambiguation required for the Ambistory task.

c) Analysis of the Ensemble Approach: An ensemble method combines the predictions of multiple models to improve overall performance. In our study, we implemented a weighted ensemble consisting of the `deberta-v3-large` encoder and a 4-bit quantized `Mistral-7B` generative model.

Using the Kaggle T4 x2 was great for speed, but each individual card only has 16GB of memory. Since our ensemble was trying to squeeze a 'Large' DeBERTa model and a '7B' Mistral model onto the same card, we kept hitting a wall where the GPU would just run out of space (OOM). To stop the code from crashing, we had to keep the training very short (10 epochs) and the batches very small. In the end, the model just didn't get enough 'practice' time to get the answers right because we were too busy fighting with the memory limits.

To avoid these crashes and stay within the compute bounds, we had to keep our batch sizes small and limit the fine-tuning of the `deberta-v3-large` model to only 10 epochs. Because larger models require more iterations to converge on complex tasks like narrative plausibility, we believe 10 epochs were insufficient for the encoder to reach its potential. This resulted in "weak" predictions that, when mixed with Mistral, failed to align with human judgments.

VI. Conclusions

In this work, we found that simpler structural prompts often outperform more complex instructional or ensemble methods for plausibility rating. Our Simple Prompt achieved the highest performance (Val : 0.53, $Test$: 0.65), while the Masked Prompt proved to be a highly competitive alternative (Val : 0.51, $Test$: 0.64) that excels in minimizing prediction error. These results demonstrate that encoder-only models like DeBERTa are highly capable of narrative understanding when provided with clear, structured inputs.

References

[1] Janosch Gehring and Michael Roth. 2025. AmbiStory: A Challenging Dataset of Lexically Ambiguous Short Stories. In Proceedings of the

14th Joint Conference on Lexical and Computational Semantics (*SEM 2025), pages 152–171, Suzhou, China. Association for Computational Linguistics.

[2] Navigli, R. (2009). Word sense disambiguation: A survey. ACM computing surveys (CSUR), 41(2), 1-69.

[3] Taher Pilehvar, M., Camacho-Collados, J. (2018). WiC: the Word-in-Context Dataset for Evaluating Context-Sensitive Meaning Representations. arXiv e-prints, arXiv-1808.

[4] Mostafazadeh, N., Chambers, N., He, X., Parikh, D., Batra, D., Vanderwende, L., ... Allen, J. (2016, June). A corpus and cloze evaluation for deeper understanding of commonsense stories. In Proceedings of the 2016 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (pp. 839-849).

[5] Teglia, S., Tedeschi, S., Navigli, R. (2025, July). How Much Do Encoder Models Know About Word Senses?. In Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers) (pp. 2266-2277).

[6] He, P., Liu, X., Gao, J., Chen, W. (2020). Deberta: Decoding-enhanced bert with disentangled attention. arXiv preprint arXiv:2006.03654.